explore all the important features of his discoveries. He must push on at high speed.

Thus it was that Hawking, in 1974, having proved firmly that a black hole radiates as though it had a temperature proportional to its surface gravity, went on to assert, without real proof, that *all* of the other similarities between the laws of black-hole mechanics and the laws of thermodynamics were more than a coincidence: The black-hole laws *are the same thing as* the thermodynamic laws, but in disguise. From this assertion and his firmly proved relationship between temperature and surface gravity, Hawking inferred a precise relationship between the hole's entropy and its surface area: The entropy is 0.10857 ... times[14] the surface area, divided by the Planck–Wheeler area. In other words, a 10-solar-mass, nonspinning hole has an entropy of 4.6×10^{78}, which is approximately the same as Bekenstein's conjecture.

Bekenstein, of course, was sure Hawking was right, and he glowed with pleasure. By the end of 1975, Zel'dovich, Starobinsky, I, and Hawking's other colleagues were also strongly inclined to agree. However, we would not feel fully satisfied until we understood the precise nature of a black hole's enormous randomness. There must be $10^{4.6 \times 10^{78}}$ ways to distribute *something* inside the black hole, without changing its external appearance (its mass, angular momentum, and charge), but what was that something? And how, in simple physical terms, could one understand the thermal behavior of a black hole—the fact that the hole behaves just like an ordinary body with temperature? As Hawking moved on to research on quantum gravity and the origin of the Universe, Paul Davies, Bill Unruh, Robert Wald, James York, I, and many others of his colleagues zeroed in on these issues. Gradually over the next ten years we arrived at the new understanding embodied in Figure 12.3.

Figure 12.3a depicts a black hole's vacuum fluctuations, as viewed by observers falling inward through the horizon. The vacuum fluctuations consist of pairs of virtual particles. Occasionally tidal gravity manages to give one of the plethora of pairs sufficient energy for its two virtual particles to become real, and for one of them to escape from the hole. This was the viewpoint on vacuum fluctuations and black-hole evaporation discussed in Figure 12.2.